

pgAdmin 4

File Object Tools Edit View Window Help

Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

No limit

Query Query History

```
1 CREATE OR REPLACE PROCEDURE insert_flight(  
2     p_flight_id INT,  
3     p_sch_departure_time TIMESTAMP,  
4     p_sch_arrival_time TIMESTAMP,  
5     p_departing_airport_id INT,  
6     p_arriving_airport_id INT,  
7     p_departing_gate TEXT,  
8     p_arriving_gate VARCHAR(50),  
9     p_airline_id INT,  
10    p_act_departure_time TIMESTAMP DEFAULT NULL,  
11    p_act_arrival_time TIMESTAMP DEFAULT NULL  
12 )  
13 LANGUAGE plpgsql  
14 AS $$  
15 BEGIN  
16     INSERT INTO flights(  
17         flight_id,  
18         sch_departure_time,  
19         sch_arrival_time,  
20         departing_airport_id,  
21         arriving_airport_id,  
22         departing_gate,  
23         arriving_gate,  
24         airline_id  
25     ) VALUES (  
26         p_flight_id,  
27         p_sch_departure_time,  
28         p_sch_arrival_time,  
29         p_departing_airport_id,  
30         p_arriving_airport_id,  
31         p_departing_gate,  
32         p_arriving_gate,  
33         p_airline_id,  
34         p_act_departure_time,  
35         p_act_arrival_time  
36     );  
37 END;  
38
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 81 msec.

Total rows: Query complete 00:00:00.081 CRLF Ln 40, Col 8

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Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
42
43
44
45 CREATE OR REPLACE PROCEDURE update_flight_times(
46     p_flight_id INT,
47     p_act_departure_time TIMESTAMP DEFAULT NULL,
48     p_act_arrival_time TIMESTAMP DEFAULT NULL
49 )
50 LANGUAGE plpgsql
51 AS $$
52 BEGIN
53     UPDATE flights
54     SET act_departure_time = COALESCE(p_act_departure_time, act_departure_time),
55         act_arrival_time = COALESCE(p_act_arrival_time, act_arrival_time),
56         updated_at = NOW()
57     WHERE flight_id = p_flight_id;
58 END $$;
59
60
61
62 CREATE OR REPLACE FUNCTION flights1_from_airport(
63     p_departing_airport_id INT
64 )
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 33 msec.

✓ Query returned successfully in 33 msec. ✕

Total rows: Query complete 00:00:00.033 CRLF Ln 45, Col 1

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Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
61
62 CREATE OR REPLACE FUNCTION flights1_from_airport(
63     p_departing_airport_id INT
64 )
65 RETURNS TABLE(
66     flight_id INT,
67     sch_departure_time TIMESTAMP,
68     sch_arrival_time TIMESTAMP,
69     departing_airport_id INT,
70     arriving_airport_id INT,
71     departing_gate TEXT,
72     arriving_gate VARCHAR(50),
73     airline_id INT,
74     act_departure_time TIMESTAMP,
75     act_arrival_time TIMESTAMP,
76     created_at TIMESTAMP,
77     updated_at TIMESTAMP
78 )
79 LANGUAGE plpgsql
80 AS $$
81 BEGIN
82     RETURN QUERY
83     SELECT *
84     FROM flights
85     WHERE departing_airport_id = p_departing_airport_id;
```

Data Output Messages Notifications

CREATE FUNCTION

Query returned successfully in 35 msec.

✓ Query returned successfully in 35 msec. ✕

Total rows: Query complete 00:00:00.035 CRLF Ln 62, Col 1

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Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
92 CREATE OR REPLACE FUNCTION avg_delay_for_airport(  
93     p_airport_id INT  
94 )  
95 RETURNS INTERVAL  
96 LANGUAGE plpgsql  
97 AS $$  
98 DECLARE  
99     v_avg_delay INTERVAL;  
100 BEGIN  
101     SELECT AVG(act_arrival_time - sch_arrival_time)  
102     INTO v_avg_delay  
103     FROM flights  
104     WHERE arriving_airport_id = p_airport_id  
105           AND act_arrival_time IS NOT NULL  
106           AND sch_arrival_time IS NOT NULL;  
107     RETURN v_avg_delay;  
108 END $$;  
109  
110  
111  
112  
113  
114  
115  
116
```

Data Output Messages Notifications

CREATE FUNCTION

Query returned successfully in 33 msec.

✓ Query returned successfully in 33 msec. ✕

Total rows: Query complete 00:00:00.033 CRLF Ln 93, Col 1

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Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
112
113
114
115
116
117
118 ✓ CREATE OR REPLACE PROCEDURE list_passengers_for_flight(
119     p_flight_id INT
120 )
121 LANGUAGE plpgsql
122 AS $$
123 BEGIN
124     SELECT p.*
125     FROM passengers p
126     JOIN booking b ON b.passenger_id = p.passenger_id
127     JOIN booking_flight bf ON bf.booking_id = b.booking_id
128     WHERE bf.flight_id = p_flight_id;
129 END $$;
130
131
132
133
134
135 ✓ CREATE OR REPLACE PROCEDURE top_passenger_by_flights()
136 LANGUAGE plpgsql
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 48 msec.

✓ Query returned successfully in 48 msec. X

Total rows: Query complete 00:00:00.048 CRLF Ln 118, Col 1

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Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
127 -- JOIN booking_flight bf ON bf.booking_id = b.booking_id
128 WHERE bf.flight_id = p_flight_id;
129 END $$;
130
131 CREATE OR REPLACE PROCEDURE top_passenger_by_flights()
132 LANGUAGE plpgsql
133 AS $$
134 BEGIN
135     SELECT p.passenger_id,
136            p.first_name,
137            p.last_name,
138            COUNT(bf.flight_id) AS flights_taken
139     FROM passengers p
140     JOIN booking b ON b.passenger_id = p.passenger_id
141     JOIN booking_flight bf ON bf.booking_id = b.booking_id
142     GROUP BY p.passenger_id, p.first_name, p.last_name
143     ORDER BY flights_taken DESC
144     LIMIT 1;
145 END $$;
146
147
148
149
150
151
152
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 31 msec.

✓ Query returned successfully in 31 msec. ✕

Total rows: Query complete 00:00:00.031 CRLF Ln 131, Col 1

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Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
127 -- JOIN booking_flight bf ON bf.booking_id = b.booking_id
128 WHERE bf.flight_id = p_flight_id;
129 END $$;
130
131 CREATE OR REPLACE PROCEDURE top_passenger_by_flights()
132 LANGUAGE plpgsql
133 AS $$
134 BEGIN
135     SELECT p.passenger_id,
136            p.first_name,
137            p.last_name,
138            COUNT(bf.flight_id) AS flights_taken
139     FROM passengers p
140     JOIN booking b ON b.passenger_id = p.passenger_id
141     JOIN booking_flight bf ON bf.booking_id = b.booking_id
142     GROUP BY p.passenger_id, p.first_name, p.last_name
143     ORDER BY flights_taken DESC
144     LIMIT 1;
145 END $$;
146
147
148
149
150
151
152
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 31 msec.

✓ Query returned successfully in 31 msec. ✕

Total rows: Query complete 00:00:00.031 CRLF Ln 131, Col 1

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File Object Tools Edit View Window Help

Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
163
164 CREATE OR REPLACE FUNCTION count_flights_for_each_airline()
165 RETURNS TABLE(
166     airline_id INT,
167     flights_count BIGINT
168 )
169 LANGUAGE plpgsql
170 AS $$
171 BEGIN
172     RETURN QUERY
173     SELECT airline_id,
174            COUNT(*) AS flights_count
175     FROM flights
176     GROUP BY airline_id;
177 END $$;
178
179
180
181
182
183
184 CREATE OR REPLACE PROCEDURE avg_ticket_price_for_flight(
185     p_flight_id INT
186 )
187 LANGUAGE plpgsql
```

Data Output Messages Notifications

CREATE FUNCTION

Query returned successfully in 33 msec.

✓ Query returned successfully in 33 msec. ✕

Total rows: Query complete 00:00:00.033 CRLF Ln 164, Col 1

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File Object Tools Edit View Window Help

Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

Query Query History

```
178
179
180
181 CREATE OR REPLACE PROCEDURE avg_ticket_price_for_flight(
182     p_flight_id INT
183 )
184 LANGUAGE plpgsql
185 AS $$
186 BEGIN
187     SELECT bf.flight_id,
188            AVG(b.ticket_price) AS avg_ticket_price
189     FROM booking_flight bf
190    JOIN booking b ON b.booking_id = bf.booking_id
191   WHERE bf.flight_id = p_flight_id
192   GROUP BY bf.flight_id;
193 END $$;
194
195
196
197
198
199
200 CREATE OR REPLACE PROCEDURE most_expensive_flight()
201 LANGUAGE plpgsql
202 AS $$
203 BEGIN
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 32 msec.

✓ Query returned successfully in 32 msec. ✕

Total rows: Query complete 00:00:00.032 CRLF Ln 181, Col 1

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Welcome db_course25/postgres@PostgreSQL 17* X

db_course25/postgres@PostgreSQL 17

No limit

Query Query History

```
187 SELECT bf.flight_id,
188       AVG(b.ticket_price) AS avg_ticket_price
189 FROM booking_flight bf
190 JOIN booking b ON b.booking_id = bf.booking_id
191 WHERE bf.flight_id = p_flight_id
192 GROUP BY bf.flight_id;
193 END $$;
194
195 CREATE OR REPLACE PROCEDURE most_expensive_flight()
196 LANGUAGE plpgsql
197 AS $$
198 BEGIN
199     SELECT f.flight_id,
200            f.departing_airport_id,
201            f.arriving_airport_id,
202            MAX(b.ticket_price) AS max_ticket_price
203 FROM flights f
204 JOIN booking_flight bf ON bf.flight_id = f.flight_id
205 JOIN booking b ON b.booking_id = bf.booking_id
206 GROUP BY f.flight_id, f.departing_airport_id, f.arriving_airport_id
207 ORDER BY max_ticket_price DESC
208 LIMIT 1;
209 END $$;
210
211
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 32 msec.

✓ Query returned successfully in 32 msec. ✕

Total rows: Query complete 00:00:00.032 CRLF Ln 195, Col 1